Tipping tests: the honest nulls

Tests for the early-warning signature of a bifurcation (B-tipping): rising autocorrelation and variance of $r(t)$ before crises, and slower recovery when the implied Kuramoto coupling is treated as the bifurcation parameter. All are null against 5,000 spectrum-preserving surrogate economy sets. In the Ashwin et al. (2012) taxonomy this points away from B-tipping and towards noise- or rate-induced transitions. A third variant (restoring time, csd_fast.py) was also null; its script and figure were lost with the September workspace.

Status key

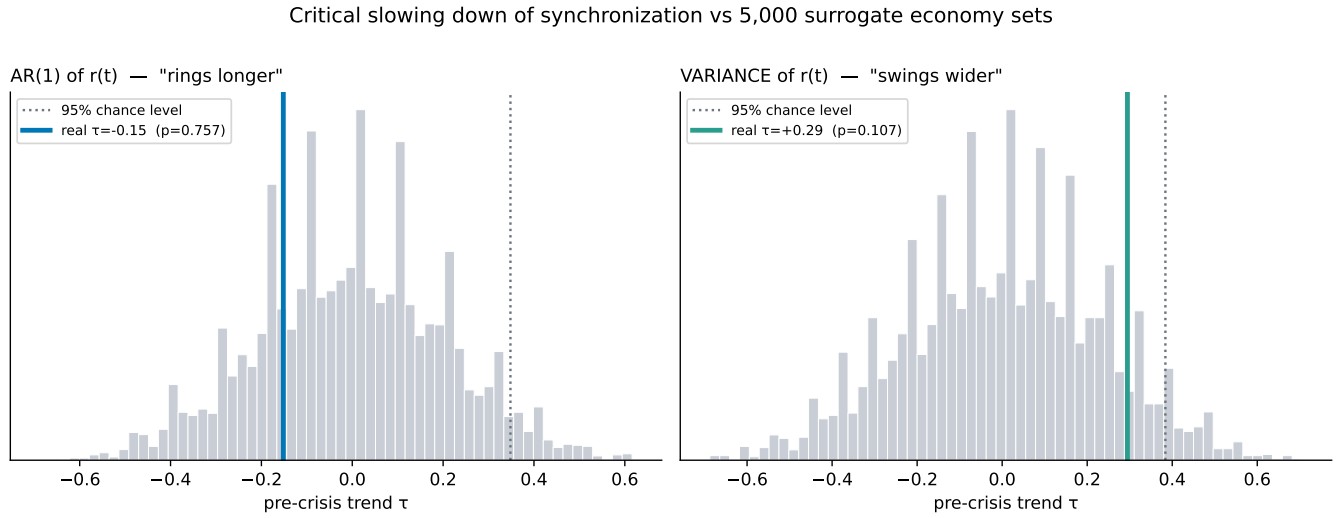
CURRENT	Paper data (Maddison 2023) and the paper's conventions; numbers match docs/methods.md.
NEEDS RELABELLING	Numbers right, but an annotation on the figure is out of date (said in the caption).
CONVENTION W	Hilbert transform applied to 1970–2022 only (convention W); the paper uses P.
SUPERSEDED	Earlier data or crisis definition; the record of how the paper got here.
DESCRIPTIVE	Atlas material from the Global Macro Database; describes, does not test.
UNVERIFIED SERIES	The series drawn matches no stored results file; redraw before use.

In this book

F6.1 Critical slowing down of synchronization: null — CURRENT
F6.2 Kuramoto coupling as the bifurcation parameter: no slowing — CURRENT

F6.1 CURRENT

Critical slowing down of synchronization: null

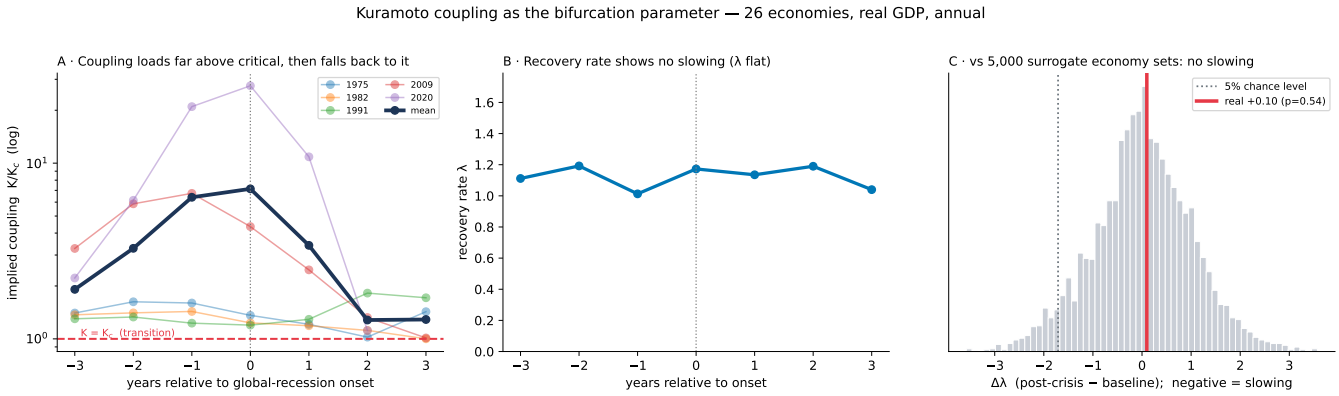


Trend (Kendall τ) of the lag-1 autocorrelation and of the variance of $r(t)$ over the 8 years before each recession (rolling 15-year windows, Gaussian detrending $\sigma = 5$), averaged over recessions, against 5,000 surrogate sets. Neither rises more than chance. Per recession, the variance trend is positive only for 2009 (+0.43) and 2020 (+0.79). Phases here come from a Hilbert transform on 1962–2022 (eight years of run-in), not the full 1950 run-in of convention P.

Numbers AR(1): τ = −0.15, p = 0.757. Variance: τ = +0.29, p = 0.107.
Made by paper1/scripts/eraA_claude_ai/csd_surrogate.py → csd_surrogate_results.json (reproduces exactly) · **Data** Maddison 2023
File figures/06_tipping_nulls/F6-01_critical_slowing_down.pdf

F6.2 CURRENT

Kuramoto coupling as the bifurcation parameter: no slowing



(A) Implied coupling $K/K_c = 1/(1-r^2)$ around each recession: far above critical before 2009 and 2020, then back to about 1. Because K/K_c is a monotone transform of r, (A) restates r rather than adding a measurement. (B) The recovery rate λ stays flat. (C) Post-crisis change in λ against 5,000 surrogates: no slowing. Phases from a Hilbert transform on 1962–2022, as in F6.1.

Numbers Mean K/K_c: 6.4 at t−1 and 7.1 at t0; $\Delta\lambda$ = +0.10, p = 0.54.
Made by paper1/scripts/eraA_claude_ai/kuramoto_bifurcation.py → kuramoto_bifurcation_results.json (reproduces exactly) · **Data** Maddison 2023
File figures/06_tipping_nulls/F6-02_kuramoto_coupling_bifurcation.pdf